

SHIJIAZHUANG, China

WMO: 536980

Lat: 38.0733N

Lon: 114.3513E

Elev: 344

StdP: 14.51

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.6	20.3	-12.4	2.8	32.8	-8.3	3.6	33.1	14.7	34.7	12.7	35.0	2.6	0	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	97.1	71.8	94.2	72.7	91.7	73.0	80.8	88.9	79.3	87.2	77.9	85.6	5.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	150.9	86.1	77.1	142.8	84.6	75.6	135.7	83.4	45.0	88.8	43.4	87.4	41.9	85.7	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.0	10.9	9.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			13.2	102.0	3.6	2.9	10.6	104.0	8.5	105.7	6.5	107.3	4.0	109.4
			11.0	83.2	3.3	1.3	8.6	84.2	6.7	84.9	4.8	85.7	2.4	86.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.6	30.4	36.6	49.1	60.9	71.9	79.8	82.4	79.7	71.8	60.1	44.9	33.8
	DBStd	19.28	5.81	6.56	7.75	7.07	6.14	5.36	4.86	4.52	5.45	6.40	7.71	5.88
	HDD50	1800	609	377	110	6	0	0	0	0	0	5	190	503
	HDD65	4296	1074	795	495	159	14	0	0	0	10	177	604	968
	CDD50	4924	0	2	83	334	677	895	1005	920	655	317	36	0
	CDD65	1944	0	0	2	37	226	445	540	455	215	24	0	0
	CDH74	19888	0	0	26	330	2018	4987	6394	4529	1462	140	2	0
	CDH80	8401	0	0	3	76	710	2411	3035	1739	408	19	0	0
Wind	WSAvg	3.9	3.5	3.8	4.6	4.9	4.8	4.1	3.7	3.5	3.3	3.3	3.3	3.5
Precipitation	PrecAvg	20.8	0.1	0.3	0.4	0.9	1.5	1.9	5.6	5.9	2.2	1.1	0.6	0.2
	PrecMax	41.3	0.5	1.6	1.7	5.1	6.4	7.4	11.1	29.6	7.0	5.6	2.3	0.8
	PrecMin	9.0	0.0	0.0	0.0	0.0	0.1	0.2	1.9	0.9	0.1	0.0	0.0	0.0
	PrecStd	6.7	0.2	0.4	0.5	1.1	1.6	1.5	2.7	5.0	1.5	1.1	0.6	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	61.1	77.2	86.8	95.7	101.1	101.5	95.9	90.9	83.1	71.7	55.5
		MCWB	39.3	43.4	52.7	61.4	64.3	69.9	71.9	76.3	71.7	61.3	50.7	39.7
	2%	DB	47.0	56.1	71.7	81.5	90.2	97.4	97.3	92.5	87.3	78.0	65.0	50.6
		MCWB	34.5	40.7	51.2	60.6	64.5	69.3	74.3	76.6	68.9	58.7	48.8	37.1
	5%	DB	43.2	51.9	66.8	77.6	87.0	94.4	94.4	90.2	84.6	74.3	60.4	46.7
		MCWB	32.3	39.1	49.4	58.4	64.2	69.5	75.3	75.9	68.3	57.4	47.3	35.3
	10%	DB	39.8	48.1	62.2	73.9	84.0	91.4	91.8	88.2	81.7	70.9	56.4	43.0
		MCWB	30.6	36.9	46.7	57.3	63.9	69.4	75.8	75.3	67.2	56.8	45.6	33.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	45.9	57.8	68.6	72.8	77.5	83.2	83.0	77.3	66.5	54.6	41.0
		MCDB	52.4	56.1	70.4	79.8	85.5	91.3	91.1	90.6	85.3	74.4	61.8	52.8
	2%	WB	35.3	42.4	53.9	64.9	70.7	75.3	81.4	80.7	74.4	63.6	51.7	38.3
		MCDB	45.2	52.8	67.7	75.4	82.8	88.7	89.8	88.2	81.7	70.4	60.9	47.7
	5%	WB	33.1	40.0	50.7	62.3	68.9	73.8	79.9	79.2	72.0	61.7	49.3	36.3
		MCDB	41.7	50.5	64.2	72.4	80.8	86.2	88.1	86.4	79.6	68.8	57.5	44.6
	10%	WB	31.1	37.7	47.9	59.7	67.0	72.4	78.6	77.8	70.0	59.9	47.2	34.5
		MCDB	38.8	47.1	60.4	69.6	78.6	84.5	86.6	84.8	77.7	67.2	54.9	41.6
Mean Daily Temperature Range	5% DB	MDBR	13.6	15.1	16.7	17.7	17.8	17.6	14.0	13.0	15.0	15.8	14.7	13.0
		MCDBR	19.2	21.6	23.3	23.9	22.9	23.4	19.2	16.1	19.1	21.1	20.9	18.7
		MCWBR	11.2	12.0	11.4	11.2	9.6	6.8	6.0	6.6	7.2	8.5	10.3	10.2
	5% WB	MCDBR	17.7	19.7	21.5	20.1	18.9	17.6	14.0	13.3	14.7	14.8	18.0	16.9
	MCWBR	10.8	11.6	11.5	11.4	9.7	6.6	6.1	6.6	7.3	7.4	9.6	9.9	
Clear-Sky Solar Irradiance	taub	0.476	0.555	0.557	0.607	0.644	0.787	0.782	0.762	0.692	0.631	0.507	0.464	
	taud	1.853	1.714	1.740	1.683	1.649	1.461	1.510	1.526	1.599	1.661	1.845	1.913	
	Ebn at Noon	204	204	223	222	217	187	187	185	187	181	193	197	
	Edh at Noon	48	63	67	75	79	96	91	87	77	66	48	43	
All-Sky Solar Radiation	RadAvg	803	998	1432	1704	1876	1696	1500	1450	1279	1048	832	733	
	RadStd	67	125	128	117	149	152	96	128	110	113	130	75	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page